University of Liberal Arts Bangladesh

Department of Computer Science & Engineering

STA 2101: Statistics & Probability

Lecture 12:
Probability Distribution

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Class Code
sdb2r4mu

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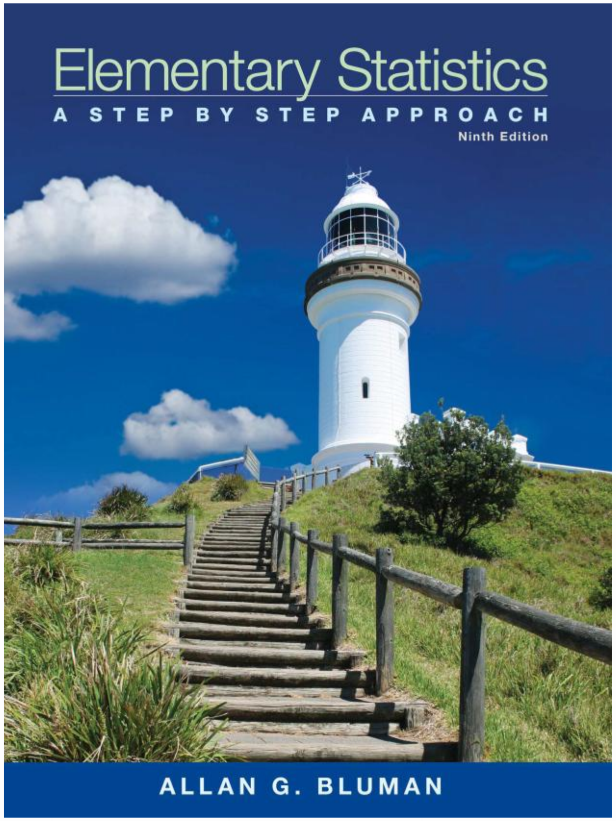
Reference Book

- **Text Book(s)**:
- Elementary Statistics – Allan G. Bluman
- Introduction to Probability and Statistics (14th Edition) By William Mendenhall, Robert J. Beaver, Barbara M. Beaver

- **Reference Material/Book(s)**:
- S Introduction to probability and statistics for engineers and scientists, Sheldon M Ross



Mark Distribution

Assessments	%
Mid Term Examination	25
Final Examination	40
Attendance and Class Participation	5
CT	5
Project	25
Total	100

Today's Topics

- Probability Distribution

Before probability distribution is defined formally, the definition of a variable is reviewed. In Chapter 1, a *variable* was defined as a characteristic or attribute that can assume different values. Various letters of the alphabet, such as X, Y, or Z, are used to represent variables. Since the variables in this chapter are associated with probability, they are called *random variables*.

For example, if a die is rolled, a letter such as X can be used to represent the outcomes. Then the value that X can assume is 1, 2, 3, 4, 5, or 6, corresponding to the outcomes of rolling a single die. If two coins are tossed, a letter, say Y, can be used to represent the number of heads, in this case 0, 1, or 2. As another example, if the temperature at 8:00 A.M. is 43° and at noon it is 53°, then the values T that the temperature assumes are said to be random, since they are due to various atmospheric conditions at the time the temperature was taken.

A **random variable** is a variable whose values are determined by chance.

Also recall from Chapter 1 that you can classify variables as discrete or continuous by observing the values the variable can assume. If a variable can assume only a specific number of values, such as the outcomes for the roll of a die or the outcomes for the toss of a coin, then the variable is called a *discrete variable*.

Discrete variables have a finite number of possible values or an infinite number of values that can be counted. The word *counted* means that they can be enumerated using the numbers 1, 2, 3, etc. For example, the number of joggers in Riverview Park each day and the number of phone calls received after a TV commercial airs are examples of discrete variables, since they can be counted.

No heads	One head			Two heads			Three heads
TTT	TTH	THT	HTT	HHT	HTH	THH	HHH
$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$
$\frac{1}{8}$	$\frac{3}{8}$			$\frac{3}{8}$			$\frac{1}{8}$

Hence, the probability of getting no heads is $\frac{1}{8}$, one head is $\frac{3}{8}$, two heads is $\frac{3}{8}$, and three heads is $\frac{1}{8}$. From these values, a probability distribution can be constructed by listing the outcomes and assigning the probability of each outcome, as shown here.

Number of heads X	0	1	2	3
Probability $P(X)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

A **discrete probability distribution** consists of the values a random variable can assume and the corresponding probabilities of the values. The probabilities are determined theoretically or by observation.

EXAMPLE 5–1 Rolling a Die

Construct a probability distribution for rolling a single die.

SOLUTION

Since the sample space is 1, 2, 3, 4, 5, 6 and each outcome has a probability of $\frac{1}{6}$, the distribution is as shown.

Outcome X	1	2	3	4	5	6
Probability $P(X)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

EXAMPLE 5–2 Tossing Coins

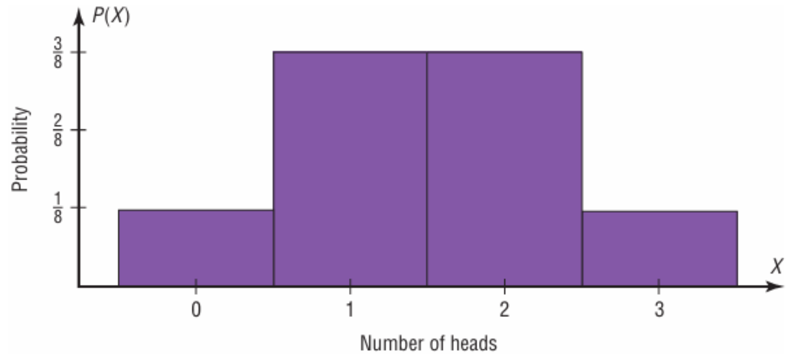
Represent graphically the probability distribution for the sample space for tossing three coins.

Number of heads X	0	1	2	3
Probability $P(X)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

SOLUTION

The values that X assumes are located on the x axis, and the values for $P(X)$ are located on the y axis. The graph is shown in Figure 5–1.

FIGURE 5–1 Probability Distribution for Example 5–2



EXAMPLE 5–3 Baseball World Series

The baseball World Series is played by the winner of the National League and that of the American League. The first team to win four games wins the World Series. In other words, the series will consist of four to seven games, depending on the individual victories. The data shown consist of 40 World Series events. The number of games played in each series is represented by the variable X. Find the probability $P(X)$ for each X, construct a probability distribution, and draw a graph for the data.

X	Number of series played
4	8
5	7
6	9
7	16
	Total 40

SOLUTION

The probability $P(X)$ can be computed for each X by dividing the number of series played for each X by the total.

For 4 games, $\frac{8}{40} = 0.200$ For 6 games, $\frac{9}{40} = 0.225$

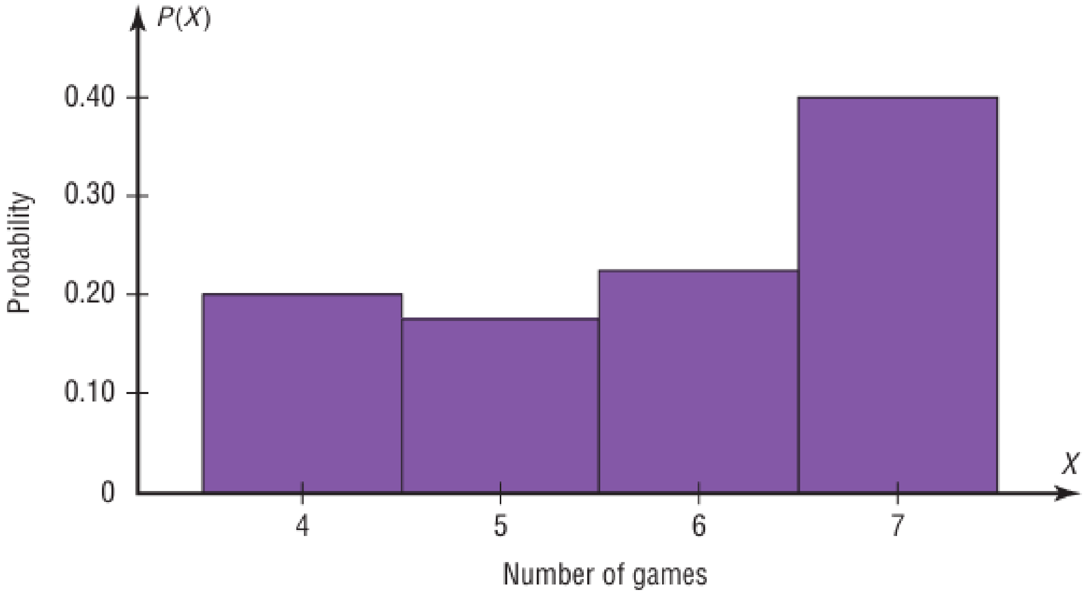
For 5 games, $\frac{7}{40} = 0.175$ For 7 games, $\frac{16}{40} = 0.400$

The probability distribution is

Number of games X	4	5	6	7
Probability $P(X)$	0.200	0.175	0.225	0.400

The graph is shown in Figure 5–2.

FIGURE 5–2
Probability Distribution for Example 5–3



Two Requirements for a Probability Distribution

1. The sum of the probabilities of all the events in the sample space must equal 1; that is, $\Sigma P(X) = 1$.
2. The probability of each event in the sample space must be between or equal to 0 and 1. That is, $0 \leq P(X) \leq 1$.

The first requirement states that the sum of the probabilities of all the events must be equal to 1. This sum cannot be less than 1 or greater than 1 since the sample space includes *all* possible outcomes of the probability experiment. The second requirement states that the probability of any individual event must be a value from 0 to 1. The reason (as stated in Chapter 4) is that the range of the probability of any individual value can be 0, 1, or any value between 0 and 1. A probability cannot be a negative number or greater than 1.

EXAMPLE 5–4 Probability Distributions

Determine whether each distribution is a probability distribution.

a.

X	5	8	11	14
$P(X)$	0.2	0.6	0.1	0.3

b.

X	1	2	3	4	5
$P(X)$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{1}{8}$	$\frac{1}{8}$

c.

X	1	2	3	4
$P(X)$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

d.

X	4	8	12
$P(X)$	-0.5	0.6	0.4

Solution

SOLUTION

a. No. The sum of the probabilities is greater than 1.

b. Yes. The sum of the probabilities of all the events is equal to 1. Each probability is greater than or equal to 0 and less than or equal to 1.

c. Yes. The sum of the probabilities of all the events is equal to 1. Each probability is greater than or equal to 0 and less than or equal to 1.

d. No. One of the probabilities is less than 0.

Mean

In Chapter 3, the mean for a sample or population was computed by adding the values and dividing by the total number of values, as shown in these formulas:

$$\text{Sample mean:} \quad \overline{X} = \frac{\Sigma X}{n} \qquad \text{Population mean:} \quad \mu = \frac{\Sigma X}{N}$$

But how would you compute the mean of the number of spots that show on top when a die is rolled? You could try rolling the die, say, 10 times, recording the number of spots, and finding the mean; however, this answer would only approximate the true mean. What about 50 rolls or 100 rolls? Actually, the more times the die is rolled, the better the approximation. You might ask, then, How many times must the die be rolled to get the exact answer? *It must be rolled an infinite number of times.* Since this task is impossible, the previous formulas cannot be used because the denominators would be infinity. Hence, a new method of computing the mean is necessary. This method gives the exact theoretical value of the mean as if it were possible to roll the die an infinite number of times.

Before the formula is stated, an example will be used to explain the concept. Suppose two coins are tossed repeatedly, and the number of heads that occurred is recorded. What will be the mean of the number of heads? The sample space is

HH, HT, TH, TT

and each outcome has a probability of $\frac{1}{4}$. Now, in the long run, you would *expect* two heads (HH) to occur approximately $\frac{1}{4}$ of the time, one head to occur approximately $\frac{1}{2}$ of the time (HT or TH), and no heads (TT) to occur approximately $\frac{1}{4}$ of the time. Hence, on average, you would expect the number of heads to be

$$\tfrac{1}{4} \cdot 2 + \tfrac{1}{2} \cdot 1 + \tfrac{1}{4} \cdot 0 = 1$$

Mean of a Probability Distribution

Formula for the Mean of a Probability Distribution

The mean of a random variable with a discrete probability distribution is

$$\mu = X_1 \cdot P(X_1) + X_2 \cdot P(X_2) + X_3 \cdot P(X_3) + \cdots + X_n \cdot P(X_n)$$
$$= \Sigma X \cdot P(X)$$

where $X_1, X_2, X_3, \ldots, X_n$ are the outcomes and $P(X_1), P(X_2), P(X_3), \ldots, P(X_n)$ are the corresponding probabilities.

Note: $\Sigma X \cdot P(X)$ means to sum the products.

EXAMPLE 5–5 Rolling a Die

Find the mean of the number of spots that appear when a die is tossed.

SOLUTION

In the toss of a die, the mean can be computed thus.

Outcome X	1	2	3	4	5	6
Probability $P(X)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

$$\mu = \Sigma X \cdot P(X) = 1 \cdot \tfrac{1}{6} + 2 \cdot \tfrac{1}{6} + 3 \cdot \tfrac{1}{6} + 4 \cdot \tfrac{1}{6} + 5 \cdot \tfrac{1}{6} + 6 \cdot \tfrac{1}{6}$$
$$= \tfrac{21}{6} = 3\tfrac{1}{2} \text{ or } 3.5$$

That is, when a die is tossed many times, the theoretical mean will be 3.5. Note that even though the die cannot show a 3.5, the theoretical average is 3.5.

The reason why this formula gives the theoretical mean is that in the long run, each outcome would occur approximately $\frac{1}{6}$ of the time. Hence, multiplying the outcome by its corresponding probability and finding the sum would yield the theoretical mean. In other words, outcome 1 would occur approximately $\frac{1}{6}$ of the time, outcome 2 would occur approximately $\frac{1}{6}$ of the time, etc.

EXAMPLE 5–6 Children in a Family

In families with five children, find the mean number of children who will be girls.

SOLUTION

First, it is necessary to find the probability distribution for the number of females. There are 32 outcomes, and there is one way for a family to have no girls. There are five ways to have one girl, that is, FMMMM, MFMMM, MMFMM, MMMFM, MMMMF. Continue with two females and three males, three females and two males, four females and one male, and five females. You can draw a tree diagram to help you.

The probability distribution is

Number of girls X	0	1	2	3	4	5
Probability $P(X)$	$\frac{1}{32}$	$\frac{5}{32}$	$\frac{10}{32}$	$\frac{10}{32}$	$\frac{5}{32}$	$\frac{1}{32}$

The mean is

$$\mu = \Sigma X \cdot P(X) = 0 \cdot \tfrac{1}{32} + 1 \cdot \tfrac{5}{32} + 2 \cdot \tfrac{10}{32} + 3 \cdot \tfrac{10}{32} + 4 \cdot \tfrac{5}{32} + 5 \cdot \tfrac{1}{32} = 2\tfrac{1}{2} = 2.5$$

Hence, the mean number of females is 2.5.

EXAMPLE 5–7 Tossing Coins

If three coins are tossed, find the mean of the number of heads that occur. (See the table preceding Example 5–1.)

SOLUTION

The probability distribution is

Number of heads X	0	1	2	3
Probability $P(X)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

The mean is

$$\mu = \Sigma X \cdot P(X) = 0 \cdot \tfrac{1}{8} + 1 \cdot \tfrac{3}{8} + 2 \cdot \tfrac{3}{8} + 3 \cdot \tfrac{1}{8} = \tfrac{12}{8} = 1\tfrac{1}{2} \text{ or } 1.5$$

The value 1.5 cannot occur as an outcome. Nevertheless, it is the long-run or theoretical average.

EXAMPLE 5–8 Number of Trips of Five Nights or More

The probability distribution shown represents the number of trips of five nights or more that American adults take per year. (That is, 6% do not take any trips lasting five nights or more, 70% take one trip lasting five nights or more per year, etc.) Find the mean.

Number of trips *X*	0	1	2	3	4
Probability *P*(*X*)	0.06	0.70	0.20	0.03	0.01

SOLUTION

$$\begin{aligned}\mu &= \Sigma X \cdot P(X) \\ &= (0)(0.06) + (1)(0.70) + (2)(0.20) + (3)(0.03) + (4)(0.01) \\ &= 0 + 0.70 + 0.40 + 0.09 + 0.04 \\ &= 1.23\end{aligned}$$

Hence, the mean of the number of trips lasting five nights or more per year taken by American adults is 1.23.

Variance and Standard Deviation

For a probability distribution, the mean of the random variable describes the measure of the so-called long-run or theoretical average, but it does not tell anything about the spread of the distribution. Recall from Chapter 3 that to measure this spread or variability, statisticians use the variance and standard deviation. These formulas were used:

$$\sigma^2 = \frac{\Sigma(X - \mu)^2}{N} \qquad \text{or} \qquad \sigma = \sqrt{\frac{\Sigma(X - \mu)^2}{N}}$$

These formulas cannot be used for a random variable of a probability distribution since N is infinite, so the variance and standard deviation must be computed differently.

To find the variance for the random variable of a probability distribution, subtract the theoretical mean of the random variable from each outcome and square the difference. Then multiply each difference by its corresponding probability and add the products. The formula is

$$\sigma^2 = \Sigma[(X - \mu)^2 \cdot P(X)]$$

Finding the variance by using this formula is somewhat tedious. So for simplified computations, a shortcut formula can be used. This formula is algebraically equivalent to the longer one and is used in the examples that follow.

Formula for the Variance of a Probability Distribution

Find the variance of a probability distribution by multiplying the square of each outcome by its corresponding probability, summing those products, and subtracting the square of the mean. The formula for the variance of a probability distribution is

$$\sigma^2 = \Sigma[X^2 \cdot P(X)] - \mu^2$$

The standard deviation of a probability distribution is

$$\sigma = \sqrt{\sigma^2} \qquad \text{or} \qquad \sigma = \sqrt{\Sigma[X^2 \cdot P(X)] - \mu^2}$$

Remember that the variance and standard deviation cannot be negative.

EXAMPLE 5–9 Rolling a Die

Compute the variance and standard deviation for the probability distribution in Example 5–5.

SOLUTION

Recall that the mean is $\mu = 3.5$, as computed in Example 5–5. Square each outcome and multiply by the corresponding probability, sum those products, and then subtract the square of the mean.

$$\sigma^2 = (1^2 \cdot \tfrac{1}{6} + 2^2 \cdot \tfrac{1}{6} + 3^2 \cdot \tfrac{1}{6} + 4^2 \cdot \tfrac{1}{6} + 5^2 \cdot \tfrac{1}{6} + 6^2 \cdot \tfrac{1}{6}) - (3.5)^2 = 2.917$$

To get the standard deviation, find the square root of the variance.

$$\sigma = \sqrt{2.917} \approx 1.708$$

Hence, the standard deviation for rolling a die is 1.708.

EXAMPLE 5–10 Selecting Numbered Balls

A box contains 5 balls. Two are numbered 3, one is numbered 4, and two are numbered 5. The balls are mixed and one is selected at random. After a ball is selected, its number is recorded. Then it is replaced. If the experiment is repeated many times, find the variance and standard deviation of the numbers on the balls.

SOLUTION

Let X be the number on each ball. The probability distribution is

Number on ball *X*	3	4	5
Probability *P*(*X*)	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{2}{5}$

The mean is

$$\mu = \Sigma X \cdot P(X) = 3 \cdot \tfrac{2}{5} + 4 \cdot \tfrac{1}{5} + 5 \cdot \tfrac{2}{5} = 4$$

The variance is

$$\begin{aligned}\sigma &= \Sigma[X^2 \cdot P(X)] - \mu^2 \\ &= 3^2 \cdot \tfrac{2}{5} + 4^2 \cdot \tfrac{1}{5} + 5^2 \cdot \tfrac{2}{5} - 4^2 \\ &= 16\tfrac{4}{5} - 16 \\ &= \tfrac{4}{5} \text{ or } 0.8\end{aligned}$$

The standard deviation is

$$\sigma = \sqrt{\frac{4}{5}} = \sqrt{0.8} \approx 0.894$$

The mean, variance, and standard deviation can also be found by using vertical columns, as shown.

X	*P*(*X*)	*X* · *P*(*X*)	$X^2 \cdot P(X)$
3	0.4	1.2	3.6
4	0.2	0.8	3.2
5	0.4	2.0	10
		$\Sigma X \cdot P(X) = 4.0$	16.8

Find the mean by summing the $\Sigma X \cdot P(X)$ column, and find the variance by summing the $X^2 \cdot P(X)$ column and subtracting the square of the mean.

$$\sigma^2 = 16.8 - 4^2 = 16.8 - 16 = 0.8$$

and

$$\sigma = \sqrt{0.8} \approx 0.894$$

EXAMPLE 5–11 On Hold for Talk Radio

A talk radio station has four telephone lines. If the host is unable to talk (i.e., during a commercial) or is talking to a person, the other callers are placed on hold. When all lines are in use, others who are trying to call in get a busy signal. The probability that 0, 1, 2, 3, or 4 people will get through is shown in the probability distribution. Find the variance and standard deviation for the distribution.

X	0	1	2	3	4
***P*(*X*)**	0.18	0.34	0.23	0.21	0.04

Should the station have considered getting more phone lines installed?

SOLUTION

The mean is

$$\begin{aligned}\mu &= \Sigma X \cdot P(X)\\ &= 0 \cdot (0.18) + 1 \cdot (0.34) + 2 \cdot (0.23) + 3 \cdot (0.21) + 4 \cdot (0.04)\\ &= 1.59\end{aligned}$$

The variance is

$$\begin{aligned}\sigma^2 &= \Sigma[X^2 \cdot P(X)] - \mu^2\\ &= [0^2 \cdot (0.18) + 1^2 \cdot (0.34) + 2^2 \cdot (0.23) + 3^2 \cdot (0.21) + 4^2 \cdot (0.04)] - 1.59^2\\ &= (0 + 0.34 + 0.92 + 1.89 + 0.64) - 2.528\\ &= 3.79 - 2.528 = 1.262\end{aligned}$$

The standard deviation is $\sigma = \sqrt{\sigma^2}$, or $\sigma = \sqrt{1.262} \approx 1.123$.

No. The mean number of people calling at any one time is 1.59. Since the standard deviation is 1.123, most callers would be accommodated by having four phone lines because $\mu + 2\sigma$ would be $1.59 + 2(1.123) = 3.836 \approx 4.0$. Very few callers would get a busy signal since at least 75% of the callers would either get through or be put on hold. (See Chebyshev's theorem in Section 3–2.)

Expectation

Another concept related to the mean for a probability distribution is that of expected value or expectation. Expected value is used in various types of games of chance, in insurance, and in other areas, such as decision theory.

The **expected value** of a discrete random variable of a probability distribution is the theoretical average of the variable. The formula is

$$\mu = E(X) = \Sigma X \cdot P(X)$$

The symbol $E(X)$ is used for the expected value.

The formula for the expected value is the same as the formula for the theoretical mean. The expected value, then, is the theoretical mean of the probability distribution. That is, $E(X) = \mu$.

When expected value problems involve money, it is customary to round the answer to the nearest cent.

EXAMPLE 5–12 Winning Tickets

One thousand tickets are sold at \$1 each for a color television valued at \$350. What is the expected value of the gain if you purchase one ticket?

SOLUTION

The problem can be set up as follows:

	Win	Lose
Gain X	\$349	−\$1
Probability $P(X)$	$\frac{1}{1000}$	$\frac{999}{1000}$

Two things should be noted. First, for a win, the net gain is \$349, since you do not get the cost of the ticket (\$1) back. Second, for a loss, the gain is represented by a negative number, in this case −\$1. The solution, then, is

$$E(X) = \$349 \cdot \frac{1}{1000} + (-\$1) \cdot \frac{999}{1000} = -\$0.65$$

Hence, a person would lose, on average, \$0.65 on each ticket purchased.

EXAMPLE 5–13 Selecting Balls

Six balls numbered 1, 2, 3, 5, 8, and 13 are placed in a box. A ball is selected at random, and its number is recorded and then it is replaced. Find the expected value of the numbers that will occur.

SOLUTION

Since the balls are replaced, the probability for each number is $\frac{1}{6}$, so the probability distribution is

Number (X)	1	2	3	5	8	13
Probability $P(X)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

The expected value is

$$E(X) = 1 \cdot \tfrac{1}{6} + 2 \cdot \tfrac{1}{6} + 3 \cdot \tfrac{1}{6} + 5 \cdot \tfrac{1}{6} + 8 \cdot \tfrac{1}{6} + 13 \cdot \tfrac{1}{6} = 5\tfrac{1}{3}$$

EXAMPLE 5–14 Bond Investment

A financial adviser suggests that his client select one of two types of bonds in which to invest \$5000. Bond X pays a return of 4% and has a default rate of 2%. Bond Y has a $2\frac{1}{2}\%$ return and a default rate of 1%. Find the expected rate of return and decide which bond would be a better investment. When the bond defaults, the investor loses all the investment.

SOLUTION

The return on bond X is $\$5000 \cdot 4\% = \$200$. The expected return then is

$$E(X) = \$200(0.98) - \$5000(0.02) = \$96$$

The return on bond Y is $\$5000 \cdot 2\frac{1}{2}\% = \$125$. The expected return then is

$$E(X) = \$125(0.99) - \$5000(0.01) = \$73.75$$

Hence, bond X would be a better investment since the expected return is higher.

Thank You

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